

Two-Level Mass–Cancellation Factorization

of Determinant Flatness:

Effective Support, Relative Phase Cancellation, and the Critical $1/J$ Threshold

Liang Wang
 School of Artificial Intelligence and Automation
 Huazhong University of Science and Technology, Wuhan 430074, P.R. China
 wangliang.f@gmail.com

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Abstract

For a finitely supported complex sequence $a = (a_n)$, put

$$Z = \sum_n a_n, \quad M = \sum_n |a_n|, \quad D = \sum_n |a_n|^2.$$

When $a \neq 0$, define the relative phase-cancellation coefficient $\mathfrak{c} = |Z|/M$, the effective support $S_{\text{eff}} = M^2/D$, and the zero-frequency concentration $\mathfrak{F} = |Z|^2/D$. We prove the exact two-level factorization

$$\boxed{\mathfrak{F} = \mathfrak{c}^2 S_{\text{eff}}}.$$

It separates magnitude spreading among coefficient bins from phase cancellation across those bins. For $a \neq 0$, we determine all elementary equality cases and prove the sharp joint strip

$$(2 - S_{\text{eff}})_+ \leq \mathfrak{F} \leq S_{\text{eff}} \leq \# \text{supp } a.$$

The upper edge is attained precisely by common-phase coefficients; the effective-support edge is attained precisely by constant modulus. The lower edge at $S_{\text{eff}} = 1$ is attained by one coefficient, its portion $1 < S_{\text{eff}} \leq 2$ by two nonzero antiparallel coefficients, and its zero portion by zero-sum configurations.

We then attach the identity to the literal fixed- h_0 TPC-32 normalized-determinant coefficient. In the natural post-bin mass regime of TPC-83, the inherited upper envelopes and the mass–energy equivalence imply

$$S_{\text{eff}}(A_X) = X^{o(1)} Q_X.$$

Consequently, at

$$Q_X = X^{267/400+o(1)}, \quad J_X = X^{133/400+o(1)},$$

the TPC-32 flatness threshold $\mathfrak{F}(A_X) \leq X^{1/400+o(1)}$ is equivalent to the relative cancellation requirement

$$\boxed{\frac{|Z_X|}{M_X} \leq \frac{X^{o(1)}}{J_X}}.$$

Since $M_X = X^{o(1)} J_X Q_X^2$ in that regime, this is also equivalent, up to subpower factors, to $|Z_X| \leq X^{o(1)} Q_X^2$. An exact exponent ledger reconciles this factorization with the TPC-82 condition $2\eta_Z \geq \lambda_D$.

The finite identities are L0, and their attachment to the literal fixed-shift coefficient is L1. No natural-mass lower bound and no new L2 cancellation estimate is proved.

Keywords: effective support; zero-frequency concentration; post-bin mass; relative cancellation; fixed shift; determinant coefficient; flatness threshold.

Convention. Every arithmetic statement concerns one prescribed nonzero h_0 , the already aggregated TPC-32 determinant coefficient, and its original global normalization. No statement is specialized to $h_0 = 2$.

1 Introduction

The high-row fixed-shift route isolates the literal determinant coefficient

$$A_X(n) = A_{\lfloor J_X \rfloor}(n)$$

only after the three matched physical channels and every native contribution in the determinant fiber have been aggregated. Its post-bin statistics are

$$Z_X = \sum_n A_X(n), \quad M_X = \sum_n |A_X(n)|, \quad D_X = \sum_n |A_X(n)|^2.$$

TPC-32 asks for an upper bound on the normalized zero-frequency concentration

$$\mathfrak{F}_X = \frac{|Z_X|^2}{D_X},$$

while TPC-83 shows that survival of the natural determinant energy is equivalent, up to subpower factors, to survival of the literal post-bin ℓ^1 mass M_X [1, 3]. These statements leave a useful question:

Once post-bin mass and energy have survived, exactly how much cancellation among determinant bins is still required?

The answer is an identity rather than an estimate. Write

$$\mathfrak{c}_X = \frac{|Z_X|}{M_X}, \quad S_{\text{eff},X} = \frac{M_X^2}{D_X}.$$

Then

$$\mathfrak{F}_X = \mathfrak{c}_X^2 S_{\text{eff},X}.$$

The two factors have different meanings. The effective support $S_{\text{eff},X}$ depends only on the magnitudes ($|A_X(n)|$); the relative cancellation $\mathfrak{c}_X$ records how the complex phases reassemble. Neither factor may be substituted for the other.

This paper makes the split exact and performs the endpoint arithmetic. Its main contributions are:

- (i) an edge-safe definition covering both nonzero and identically zero complex sequences;
- (ii) the exact factorization $\mathfrak{F} = \mathfrak{c}^2 S_{\text{eff}}$;
- (iii) sharp ranges and complete elementary equality cases;
- (iv) for a nonzero coefficient, a joint lower edge $(2 - S_{\text{eff}})_+ \leq \mathfrak{F}$, showing what very concentrated magnitude profiles force even under optimal phase opposition;
- (v) exact counterexamples showing that identical magnitude data can produce maximal or complete zero-mode cancellation;
- (vi) a literal crosswalk to the fixed- h_0 TPC-32 coefficient;

- (vii) a proof that natural post-bin mass implies $S_{\text{eff},X} = X^{o(1)}Q_X$;
- (viii) the critical equivalence

$$\mathfrak{F}_X \leq X^{1/400+o(1)} \iff \frac{|Z_X|}{M_X} \leq \frac{X^{o(1)}}{J_X};$$

and

- (ix) a two-level exponent ledger that reproduces, without double counting, the TPC-82 compatibility condition $2\eta_Z \geq \lambda_D$.

The factorization does not create phase cancellation. It identifies its exact relative scale after the magnitude-spreading level has been fixed.

1.1 Audit levels

We retain the established three-level terminology. An L0 statement is finite algebra or an abstract finite counterexample. An L1 statement identifies the same quantities on the literal fixed- h_0 carrier with its original aggregation and normalization. An L2 statement is a new growing- X arithmetic estimate for that carrier. The finite theorems in [Sections 2 and 3](#) are L0. The crosswalk and conditional exponent implications are L1. No L2 input is asserted.

2 Exact two-level factorization

Let $a = (a_n)_{n \in \mathbb{Z}}$ be finitely supported and complex. Define

$$S = \#\text{supp } a, \quad L = \|a\|_\infty, \quad M = \|a\|_1, \quad D = \|a\|_2^2, \quad Z = \sum_n a_n. \quad (1)$$

2.1 The zero sequence

Lemma 2.1 (Edge equivalence). *The following are equivalent:*

$$a = 0, \quad M = 0, \quad D = 0.$$

Whenever they hold, $S = L = Z = 0$.

Proof. Both M and D are sums of nonnegative terms and vanish exactly when every coefficient vanishes. The remaining conclusions are then immediate. $\square$

For $a = 0$, ratios such as M^2/D and $|Z|/M$ are formally $0/0$. We remove the ambiguity by convention:

$$\mathfrak{c}(0) = 0, \quad S_{\text{eff}}(0) = 0, \quad \mathfrak{F}(0) = 0. \quad (2)$$

This agrees with the zero-flatness convention used in the physical route. All division in the remainder of the finite theory is restricted to $a \neq 0$.

2.2 The factorization

Definition 2.2 (Two levels). For $a \neq 0$, define

$$\mathfrak{c}(a) = \frac{|Z|}{M}, \quad S_{\text{eff}}(a) = \frac{M^2}{D}, \quad \mathfrak{F}(a) = \frac{|Z|^2}{D}. \quad (3)$$

We call $\mathfrak{c}$ the relative phase-cancellation coefficient, S_{eff} the effective support, and $\mathfrak{F}$ the zero-frequency concentration.

Theorem 2.3 (Exact mass–cancellation factorization). *For every finitely supported complex sequence,*

$$\boxed{\mathfrak{F}(a) = \mathfrak{c}(a)^2 S_{\text{eff}}(a)}. \quad (4)$$

For $a \neq 0$, the elementary ranges are

$$0 \leq \mathfrak{c}(a) \leq 1, \quad 1 \leq S_{\text{eff}}(a) \leq S, \quad 0 \leq \mathfrak{F}(a) \leq S_{\text{eff}}(a) \leq S. \quad (5)$$

All three statistics are invariant under multiplication of a by a nonzero complex scalar.

Proof. For $a = 0$, (4) is $0 = 0^2 \cdot 0$ by convention. Suppose $a \neq 0$. Then $M, D > 0$, and direct multiplication gives

$$\mathfrak{c}^2 S_{\text{eff}} = \frac{|Z|^2 M^2}{M^2 D} = \frac{|Z|^2}{D} = \mathfrak{F}.$$

The triangle inequality gives $|Z| \leq M$, hence $0 \leq \mathfrak{c} \leq 1$. Also

$$D \leq M^2$$

and Cauchy–Schwarz on the exact support gives

$$M^2 \leq SD.$$

Thus $1 \leq S_{\text{eff}} \leq S$. The remaining bounds follow from the factorization. Each numerator and denominator has the same homogeneity needed for scalar invariance. $\square$

Remark 2.4 (Magnitude level versus phase level). The statistic S_{eff} depends only on $(|a_n|)$. For fixed magnitudes, changing phases leaves S_{eff} , M , and D unchanged but can change $\mathfrak{c}$ and $\mathfrak{F}$. This is the exact sense in which the factorization has two levels.

2.3 Equality cases

Theorem 2.5 (Equality dictionary). *Let $a \neq 0$.*

(i) $\mathfrak{c} = 1$ if and only if all nonzero coefficients lie on one closed complex ray: there is $\zeta \in \mathbb{C}$, $|\zeta| = 1$, such that

$$a_n = \zeta |a_n| \quad (n \in \text{supp } a).$$

(ii) $\mathfrak{c} = 0$ and $\mathfrak{F} = 0$ if and only if $Z = 0$.

(iii) $S_{\text{eff}} = S$ if and only if $|a_n|$ is constant on $\text{supp } a$.

(iv) $S_{\text{eff}} = 1$ if and only if $S = 1$.

(v) $\mathfrak{F} = S_{\text{eff}}$ if and only if the common-ray condition in (i) holds.

(vi) $\mathfrak{F} = S$ if and only if a_n is one fixed nonzero complex number on $\text{supp } a$.

Proof. Part (i) is the equality case of the complex triangle inequality. Part (ii) follows from $D, M > 0$. Part (iii) is the equality case of

$$\left(\sum_{n \in \text{supp } a} |a_n| \right)^2 \leq S \sum_{n \in \text{supp } a} |a_n|^2.$$

For (iv), expand

$$M^2 = D + 2 \sum_{m < n} |a_m| |a_n|.$$

Equality $M^2 = D$ holds exactly when at most one coefficient is nonzero. Parts (v) and (vi) follow from $\mathfrak{F} = \mathfrak{c}^2 S_{\text{eff}}$: equality in (v) requires $\mathfrak{c} = 1$, while equality $\mathfrak{F} = S$ requires both $\mathfrak{c} = 1$ and $S_{\text{eff}} = S$. The latter two conditions say that all moduli and all phases agree. $\square$

3 A sharp joint range and phase countermodels

The factorization gives the upper edge $\mathfrak{F} \leq S_{\text{eff}}$. There is also a universal lower edge when the magnitude profile is too concentrated to cancel completely.

Theorem 3.1 (Sharp effective-support strip). *For every nonzero finitely supported complex sequence,*

$$\boxed{(2 - S_{\text{eff}})_+ \leq \mathfrak{F} \leq S_{\text{eff}}}. \quad (6)$$

The upper equality is characterized by [Theorem 2.5](#). If $S_{\text{eff}} < 2$, equality on the lower edge occurs exactly in the following cases:

- (a) $S = 1$; or
- (b) $S = 2$ and the two coefficients are antiparallel.

If $S_{\text{eff}} \geq 2$, equality on the displayed zero lower edge occurs exactly when $Z = 0$.

Proof. Writing the support as a finite ordered set,

$$\begin{aligned} |Z|^2 &= D + 2 \sum_{m < n} \operatorname{Re}(a_m \overline{a_n}) \\ &\geq D - 2 \sum_{m < n} |a_m| |a_n| \\ &= D - (M^2 - D) = 2D - M^2. \end{aligned}$$

Division by D gives $\mathfrak{F} \geq 2 - S_{\text{eff}}$; nonnegativity gives the positive part. The upper edge is already in [Theorem 2.3](#).

When $S_{\text{eff}} < 2$, the lower right side is positive. Equality requires

$$\operatorname{Re}(a_m \overline{a_n}) = -|a_m| |a_n|$$

for every pair of nonzero coefficients. Thus every pair must be antiparallel. This is possible for one coefficient, vacuously, or for two coefficients, and is impossible for three nonzero complex numbers. Conversely, one coefficient or two antiparallel coefficients gives equality. When $S_{\text{eff}} \geq 2$, the displayed lower edge is zero, so equality there is equivalent to $\mathfrak{F} = 0$, hence to $Z = 0$. $\square$

Proposition 3.2 (Sharpness of the lower edge up to its zero endpoint). *Every value $s_e \in [1, 2]$ occurs on the lower edge of (6); the endpoint $s_e = 2$ is the zero endpoint. More precisely, $s_e = 1$ is attained by one nonzero coefficient, while for every $1 < s_e \leq 2$ there are two nonzero antiparallel coefficients for which*

$$S_{\text{eff}} = s_e, \quad \mathfrak{F} = 2 - s_e.$$

Proof. For $s_e = 1$, take the one-point sequence $a = (1)$. For the remaining values, take $a = (1, -t)$ with $0 < t \leq 1$. Then

$$S_{\text{eff}}(t) = \frac{(1+t)^2}{1+t^2}, \quad \mathfrak{F}(t) = \frac{(1-t)^2}{1+t^2} = 2 - S_{\text{eff}}(t).$$

The continuous function $S_{\text{eff}}(t)$ increases from 1 to 2. $\square$

Proposition 3.3 (Pointwise sharpness of both edges). *For every $s_e \geq 1$, some nonzero finite sequence satisfies*

$$S_{\text{eff}} = s_e, \quad \mathfrak{F} = S_{\text{eff}}.$$

For every $s_e \geq 2$, some nonzero finite sequence satisfies

$$S_{\text{eff}} = s_e, \quad \mathfrak{F} = 0.$$

Thus both pieces of the boundary in (6) are pointwise attainable.

Proof. Choose an integer $k \geq s_e$. For one coefficient of magnitude 1 and $k - 1$ coefficients of common magnitude $t \in [0, 1]$, all with the same phase, the effective support is

$$E_k(t) = \frac{(1 + (k - 1)t)^2}{1 + (k - 1)t^2}.$$

It increases continuously from 1 to k , and the common-phase condition gives $\mathfrak{F} = S_{\text{eff}}$.

For the zero edge, choose k with $2k \geq s_e$ and use opposite pairs with magnitudes

$$(1, 1, t, t, \dots, t, t).$$

Their signed coefficients may be written

$$(1, -1, t, -t, \dots, t, -t).$$

They sum to zero, while their effective support is

$$2E_k(t),$$

which increases continuously from 2 to $2k$. Hence every $s_e \geq 2$ occurs with $\mathfrak{F} = 0$. $\square$

Example 3.4 (Same magnitude level, opposite phase outcomes). Let $S \geq 2$ be even and $L > 0$. On a support of size S , let

$$a_n = L$$

for every n , and let b_n consist of $S/2$ values L and $S/2$ values $-L$. Both sequences have

$$M = SL, \quad D = SL^2, \quad S_{\text{eff}} = S.$$

But

$$\mathfrak{c}(a) = 1, \quad \mathfrak{F}(a) = S, \quad \mathfrak{c}(b) = 0, \quad \mathfrak{F}(b) = 0.$$

Thus even exact natural effective support does not imply any phase cancellation.

Remark 3.5 (Complex phases cause no hidden hypothesis). All proofs are over $\mathbb{C}$. The terms “common phase” and “antiparallel” are equality conditions in the complex triangle inequality, not assumptions that the physical coefficient is real.

4 Literal fixed-shift crosswalk

Fix one nonzero integer h_0 . On the high- β TPC-32 row, put

$$Q_X = X^{267/400+o(1)}, \quad J_X = X^{133/400+o(1)}, \quad C_X = \lfloor J_X \rfloor, \quad (7)$$

and

$$N_{0,X} = J_X Q_X^2. \quad (8)$$

Let

$$A_X(n) = A_{C_X}(n), \quad n \in \mathbb{Z},$$

be the literal signed normalized-determinant coefficient after all three matched channels and all native contributions in each determinant bin have been summed. Define

$$\begin{aligned} S_X &= \# \text{supp } A_X, & L_X &= \|A_X\|_\infty, \\ M_X &= \|A_X\|_1, & D_X &= \|A_X\|_2^2, \\ Z_X &= \sum_n A_X(n). \end{aligned} \quad (9)$$

If $A_X = 0$, use the convention

$$\mathfrak{c}_X = S_{\text{eff},X} = \mathfrak{F}_X = 0.$$

Otherwise put

$$\mathfrak{c}_X = \frac{|Z_X|}{M_X}, \quad S_{\text{eff},X} = \frac{M_X^2}{D_X}, \quad \mathfrak{F}_X = \frac{|Z_X|^2}{D_X}. \quad (10)$$

Proposition 4.1 (Imported TPC-32 envelopes). *For the coefficient above,*

$$S_X \ll Q_X, \quad (11)$$

$$M_X \ll X^{o(1)} N_{0,X}, \quad (12)$$

$$L_X \ll X^{o(1)} \frac{N_{0,X}}{Q_X}, \quad (13)$$

$$D_X \ll X^{o(1)} \frac{N_{0,X}^2}{Q_X}. \quad (14)$$

Every quantity is post-aggregation and uses the original global normalization.

Proof. At $C_X = \lfloor J_X \rfloor$, one has $C_X \leq J_X \ll Q_X$, so the support, post-fiber majorant, singleton, and energy estimates of TPC-32 apply [1]. Their literal specialization is exactly (11)–(14), as recorded in TPC-83 [3]. $\square$

Theorem 4.2 (Literal two-level identity). *For every X ,*

$$\boxed{\mathfrak{F}_X = \mathfrak{c}_X^2 S_{\text{eff},X}}. \quad (15)$$

When $A_X \neq 0$,

$$(2 - S_{\text{eff},X})_+ \leq \mathfrak{F}_X \leq S_{\text{eff},X} \leq S_X \ll Q_X. \quad (16)$$

Proof. Apply Theorems 2.3 and 3.1 to A_X . The zero case follows from the common convention, and the final support bound is (11). $\square$

Remark 4.3 (Post-bin means post-bin). The mass M_X in (10) is

$$\sum_n |A_X(n)|,$$

after cancellation within every determinant bin. A raw atomic mass, the absolute TPC-32 majorant, a pre-bin mass, or a sum of separately renormalized channel masses may upper-bound M_X , but cannot replace it in (15). TPC-83 gives exact finite counterexamples to every such reverse substitution [3].

Remark 4.4 (Broad support is not yet effective support). The estimate $S_X \asymp Q_X$, even if independently known, would not imply $S_{\text{eff},X} \asymp Q_X$: the magnitude mass could be concentrated on a much smaller subset. The effective-support conclusion below uses the literal mass and energy scales, not support cardinality alone.

5 Natural mass and the critical relative-cancellation threshold

Definition 5.1 (Natural post-bin mass regime). The literal coefficient is in the natural post-bin mass regime if

$$M_X \geq X^{-o(1)} N_{0,X}. \quad (17)$$

Together with (12), this means

$$M_X = X^{o(1)} N_{0,X},$$

where the notation denotes two-sided comparability up to subpower factors.

TPC-83 proves that (17) is equivalent to natural determinant-energy survival:

$$D_X \geq X^{-o(1)} \frac{N_{0,X}^2}{Q_X}. \quad (18)$$

Combined with (14), this gives two-sided subpower comparability for D_X .

Theorem 5.2 (Natural mass forces natural effective support). *In the natural post-bin mass regime,*

$$\boxed{S_{\text{eff},X} = X^{o(1)} Q_X.} \quad (19)$$

Equivalently,

$$X^{-o(1)} Q_X \leq S_{\text{eff},X} \leq X^{o(1)} Q_X.$$

Consequently,

$$S_X \geq S_{\text{eff},X} \geq X^{-o(1)} Q_X, \quad (20)$$

recovering the broad-actual-support conclusion of TPC-83.

Proof. Use the mass lower bound and the energy upper bound:

$$S_{\text{eff},X} = \frac{M_X^2}{D_X} \geq \frac{X^{-o(1)} N_{0,X}^2}{X^{o(1)} N_{0,X}^2 / Q_X} = X^{-o(1)} Q_X.$$

For the reverse inequality, use the mass upper bound and the TPC-83 energy lower bound:

$$S_{\text{eff},X} \leq \frac{X^{o(1)} N_{0,X}^2}{X^{-o(1)} N_{0,X}^2 / Q_X} = X^{o(1)} Q_X.$$

Finally, $S_{\text{eff},X} \leq S_X$ by Theorem 2.3. □

Remark 5.3 (No phase information has been gained). Theorem 5.2 fixes the magnitude-spreading level up to subpower factors. It gives no upper bound for $\mathfrak{c}_X$. The common-phase and balanced-phase examples in Example 3.4 both have maximal effective support.

5.1 A general threshold conversion

Theorem 5.4 (Effective-support to cancellation conversion). *Suppose, for fixed exponents $\sigma, \chi \geq 0$, that*

$$S_{\text{eff},X} = X^{\sigma+o(1)}.$$

Then

$$\boxed{\mathfrak{F}_X \leq X^{\chi+o(1)} \iff \mathfrak{c}_X \leq X^{(\chi-\sigma)/2+o(1)}.} \quad (21)$$

When $\sigma > \chi$, the right side is a genuine relative cancellation requirement.

Proof. The exact factorization gives

$$\mathfrak{c}_X^2 = \frac{\mathfrak{F}_X}{S_{\text{eff},X}}.$$

Insert the two-sided asymptotic for $S_{\text{eff},X}$ in either direction and take square roots. □

Theorem 5.5 (Critical $1/J$ equivalence). *Assume the natural post-bin mass regime at*

$$\beta = \frac{267}{400}.$$

Then the TPC-32 flatness threshold and the relative cancellation requirement are equivalent:

$$\boxed{\begin{aligned} \mathfrak{F}_X \leq X^{1/400+o(1)} &\iff \mathfrak{c}_X = \frac{|Z_X|}{M_X} \\ &\leq X^{-133/400+o(1)} = \frac{X^{o(1)}}{J_X}. \end{aligned}} \quad (22)$$

Proof. By [Theorem 5.2](#),

$$S_{\text{eff},X} = X^{267/400+o(1)}.$$

Apply [Theorem 5.4](#) with $\sigma = 267/400$ and $\chi = 1/400$:

$$\frac{\sigma - \chi}{2} = \frac{1}{2} \left(\frac{267}{400} - \frac{1}{400} \right) = \frac{133}{400}.$$

The critical scale relation gives

$$J_X = X^{133/400+o(1)}.$$

□

Corollary 5.6 (Relative and absolute zero-mode targets). *In the same natural mass regime, the following three statements are equivalent up to subpower factors:*

$$\mathfrak{F}_X \leq X^{1/400+o(1)}, \quad (23)$$

$$\frac{|Z_X|}{M_X} \leq \frac{X^{o(1)}}{J_X}, \quad (24)$$

$$|Z_X| \leq X^{o(1)} Q_X^2. \quad (25)$$

Proof. The first two are equivalent by [Theorem 5.5](#). Natural mass gives

$$X^{-o(1)} J_X Q_X^2 \leq M_X \leq X^{o(1)} J_X Q_X^2.$$

Multiplying (24) by the upper bound for M_X gives (25). Conversely, divide (25) by the lower bound for M_X . □

Remark 5.7 (Why the factor is $1/J$). The $1/J_X$ is not inserted heuristically. It is the square-root gap between the natural effective-support exponent $267/400$ and the allowed flatness exponent $1/400$:

$$\frac{1}{2} \left(\frac{267}{400} - \frac{1}{400} \right) = \frac{133}{400}.$$

6 The exact two-level exponent ledger

The natural regime is the zero-loss corner of a more general conditional ledger. The purpose of this section is bookkeeping: none of the displayed exponent profiles is proved for the physical coefficient.

Definition 6.1 (Two-sided exponent profile). Assume $A_X \neq 0$ for all sufficiently large X . Let

$$Q_X = X^{\beta+o(1)}, \quad J_X = X^{1-\beta+o(1)}, \quad N_{0,X} = J_X Q_X^2.$$

A two-sided profile (μ, λ, η_c) , with $\mu, \lambda, \eta_c \geq 0$, means

$$M_X = X^{-\mu+o(1)} N_{0,X}, \quad (26)$$

$$D_X = X^{-\lambda+o(1)} \frac{N_{0,X}^2}{Q_X}, \quad (27)$$

$$\mathfrak{c}_X = X^{-\eta_c+o(1)}. \quad (28)$$

The exponent η_c measures relative cancellation against the actual post-bin mass. A finite profile of this form forces $Z_X \neq 0$ eventually. If $Z_X = 0$ on a subsequence, then $\mathfrak{F}_X = 0$ there and every flatness upper bound is automatic; no finite η_c or η_Z is assigned on that zero-mode subsequence.

Theorem 6.2 (Profile identities). *Under Definition 6.1,*

$$S_{\text{eff},X} = X^{\beta+\lambda-2\mu+o(1)}, \quad (29)$$

$$\mathfrak{F}_X = X^{\beta+\lambda-2\mu-2\eta_\epsilon+o(1)}. \quad (30)$$

The TPC-32 bound

$$\mathfrak{F}_X \leq X^{\chi+o(1)}$$

is certified by the profile exactly when

$$\boxed{2\eta_\epsilon \geq \beta - \chi + \lambda - 2\mu.} \quad (31)$$

Proof. Divide the square of (26) by (27):

$$S_{\text{eff},X} = X^{-2\mu+\lambda+o(1)} Q_X,$$

which is (29). Multiply by $\mathfrak{c}_X^2$ and use the exact factorization to obtain (30). Comparing its exponent with χ gives (31). $\square$

Proposition 6.3 (Feasible mass-energy wedge). *Any two-sided profile compatible with the verified TPC-32 envelopes satisfies*

$$\boxed{\mu \leq \lambda \leq 2\mu.} \quad (32)$$

Consequently,

$$\beta - \mu \leq \beta + \lambda - 2\mu \leq \beta,$$

subject also to the trivial nonnegative effective-support exponent.

Proof. The lower mass-to-energy inequality and $S_X \ll Q_X$ give

$$D_X \gg X^{-2\mu+o(1)} \frac{N_{0,X}^2}{Q_X}.$$

Comparison with (27) yields $\lambda \leq 2\mu$. Conversely,

$$M_X \geq \frac{D_X}{L_X} \gg X^{-\lambda+o(1)} N_{0,X}$$

by (13). Comparison with (26) yields $\mu \leq \lambda$. The effective-support bounds follow from (29). $\square$

6.1 Reconciliation with the absolute zero-mode ledger

Define the signed absolute zero-mode exponent η_Z by

$$|Z_X| = X^{-\eta_Z+o(1)} Q_X^2. \quad (33)$$

It may be negative if the zero mode is larger than Q_X^2 .

Proposition 6.4 (Relative-to-absolute exponent dictionary). *Under the two-sided profile,*

$$\boxed{\eta_Z = \eta_\epsilon + \mu - (1 - \beta).} \quad (34)$$

At the critical TPC-32 endpoint

$$\chi = \frac{1}{400}, \quad \beta = \frac{267}{400},$$

the relative ledger (31) is exactly

$$\boxed{2\eta_Z \geq \lambda.} \quad (35)$$

Proof. From $|Z_X| = \mathfrak{c}_X M_X$,

$$|Z_X| = X^{-\eta_\kappa - \mu + o(1)} J_X Q_X^2 = X^{-\eta_\kappa - \mu + 1 - \beta + o(1)} Q_X^2,$$

which proves (34). At the critical endpoint,

$$\beta - \chi = \frac{266}{400} = 2(1 - \beta).$$

Substitute $\eta_\kappa = \eta_Z - \mu + (1 - \beta)$ into (31); all mass terms cancel, leaving $2\eta_Z \geq \lambda$. $\square$

Corollary 6.5 (Natural corner). *At $\mu = \lambda = 0$, the critical relative requirement is*

$$\eta_\kappa \geq \frac{133}{400} = 1 - \beta,$$

equivalently

$$\mathfrak{c}_X \leq \frac{X^{o(1)}}{J_X}.$$

Proof. Insert $\mu = \lambda = 0$ and $\beta - \chi = 266/400$ into (31). $\square$

Remark 6.6 (No duplicated gain). The mass exponent μ appears once in the effective-support level and once, with the opposite bookkeeping role, in the conversion from relative to absolute cancellation. Equation (35) shows that it cancels exactly. Assigning the same mass gain or loss again in the TPC-82 absolute ledger would double count it [2].

Definition 6.7 (Independent physical-loss ledger). Let Λ_{phys} be the sum of fixed-power losses actually incurred by the full physical interface and downstream comparison, excluding any loss already represented in $\lambda - 2\eta_Z$. The route continues only under the strict rule

$$\Lambda_{\text{phys}} < \frac{1}{400}.$$

Equality or excess is a stop condition. This ledger is independent of (35); the same loss is never charged in both.

7 Claim boundary and operational consequences

Theorem 7.1 (Audited conclusion). *For every finitely supported complex coefficient,*

$$\mathfrak{F} = \mathfrak{c}^2 S_{\text{eff}}.$$

For every nonzero coefficient,

$$(2 - S_{\text{eff}})_+ \leq \mathfrak{F} \leq S_{\text{eff}} \leq S.$$

For the literal TPC-32 determinant coefficient at one fixed nonzero h_0 , natural post-bin mass survival implies

$$S_{\text{eff},X} = X^{o(1)} Q_X.$$

At

$$\beta = \frac{267}{400},$$

the TPC-32 condition

$$\mathfrak{F}_X \leq X^{1/400 + o(1)}$$

is then equivalent to

$$\frac{|Z_X|}{M_X} \leq \frac{X^{o(1)}}{J_X},$$

and to the absolute target

$$|Z_X| \leq X^{o(1)} Q_X^2.$$

Proof. Combine Theorems 2.3, 3.1, 5.2 and 5.5 and Corollary 5.6. $\square$

What is proved. The factorization, sharp strip, equality cases, and finite countermodels are unconditional L0 results. Their identification on the already aggregated fixed- h_0 coefficient, together with the conditional natural-mass and exponent transfers, is an L1 result.

What is not proved. This paper proves no lower bound for M_X or D_X , no upper bound for $\mathfrak{c}_X$ or Z_X , and no new fixed-shift Möbius-correlation estimate. It proves no parity breakthrough, no Hardy–Littlewood asymptotic, no prime-pair lower bound, and no twin-prime result. Nothing is specialized from a general nonzero h_0 to $h_0 = 2$.

Edge-case rule. If $A_X = 0$, then $M_X = D_X = Z_X = 0$ and the convention $\mathfrak{c}_X = S_{\text{eff},X} = \mathfrak{F}_X = 0$ preserves the exact factorization and the nonnegative upper bounds. The lower strip $(2 - S_{\text{eff},X})_+ \leq \mathfrak{F}_X$ is asserted only for $A_X \neq 0$. This trivial flatness does not certify natural mass or energy survival. No expression containing M_X^{-1} or D_X^{-1} may be used on that branch.

Normalization rule. Only the post-bin mass

$$M_X = \sum_n |A_X(n)|$$

enters the factorization. A raw absolute mass or TPC-32 majorant cannot be substituted for it. Likewise, actual support $S_X \asymp Q_X$ would not by itself imply $S_{\text{eff},X} \asymp Q_X$.

Exponent rule. The shorthand $1/J_X$ is valid at the critical endpoint only in the natural mass–energy corner. With nonzero losses, the correct relative requirement is

$$2\eta_{\mathfrak{c}} \geq \beta - \chi + \lambda - 2\mu.$$

At $\chi = 1/400$ this is equivalent to the absolute ledger $2\eta_Z \geq \lambda$. Neither the geometric $1/400$ baseline nor any mass exponent may be spent twice.

Independent physical-loss rule. The compatibility $2\eta_Z \geq \lambda$ is not the full-route loss budget. Separately, every fixed-power loss actually entering the physical interface and downstream comparison, but not already represented in $\lambda - 2\eta_Z$, must satisfy

$$\Lambda_{\text{phys}} < \frac{1}{400}.$$

Equality or excess stops the route, and no loss is charged in both ledgers.

Next gate. After natural post-bin mass is established, the remaining arithmetic task is exactly a relative phase-cancellation theorem at scale

$$\frac{|Z_X|}{M_X} \ll \frac{X^{o(1)}}{J_X}.$$

Without natural mass, the full profile $(\mu, \lambda, \eta_{\mathfrak{c}})$ must be measured first. A numerical observation, averaged-shift result, or generic-frequency estimate may motivate that theorem but is not an L2 substitute for the literal fixed- h_0 bound.

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